

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

- For water (a), calculate the heat change due to temperature change .

- Use the formula : $q = m * c * \Delta T$.

- For a 1 liter of water (1000 g), ΔT is 30°C .

- For aluminum (b), use the same formula , $m = 1000\text{ g}$, $c = 0.215\text{ cal/g}^\circ\text{C}$, ΔT is 30°C .

</think>

<answer>

A

</answer>